

Titration report



Method

Method ID	SERCALKSAMPLES	Release state	Released
Name	TA of Seawater	Modified by	SystemInternal
Modified on	2/22/2024 11:07:08 AM		

Summary

Samples

	Sample ID	Name	Result	State
1	FTS2023GW-TS1NW1 8/16 11:00rep1, 2	Sample size	51.964 g	
		Content R1	NaN mmol/L	Not OK
2	FTS2023GW-TS1NW1 8/16 11:00rep2, 3	Sample size	42.547 g	
		Content R1	NaN mmol/L	Not OK
3	FTS2023GW-T2NW1 8/16 12:00rep1, 4	Sample size	50.991 g	
		Content R1	NaN mmol/L	Not OK
4	FTS2023GW-T2NW1 8/16 12:00rep2, 5	Sample size	59.02 g	
		Content R1	NaN mmol/L	Not OK

Statistics

R1

Minimum	NaN mmol/L	Minimum	NaN mmol/L
Standard deviation	NaN	Rel. standard deviation	NaN %
Total samples	4	Samples excluded	0

Main Result Details

Result title	Content	Sample ID1	FTS2023GW-TS1NW1 8/16
Result name	R1		11:00rep1
Result	NaN mmol/L	Sample ID2	2
		Result comment	
		Result state	Not OK
		Result state comment	
		Nominal value	
		Compliance state	Released
		Is result deleted	No

Calculation formula R1=Q*C/m

Result title	Content	Sample ID1	FTS2023GW-TS1NW1 8/16
Result name	R1		11:00rep2
Result	NaN mmol/L	Sample ID2	3
		Result comment	
		Result state	Not OK
		Result state comment	
		Nominal value	

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Compliance state
Is result deleted

Released
No

Calculation formula R1=Q*C/m

Result title Content

Result name R1

Result NaN mmol/L

Sample ID1 FTS2023GW-T2NW1 8/16

12:00rep1

Sample ID2 4

Result comment

Result state Not OK

Result state comment

Nominal value

Compliance state Released

Is result deleted No

Calculation formula R1=Q*C/m

Result title Content

Result name R1

Result NaN mmol/L

Sample ID1 FTS2023GW-T2NW1 8/16

12:00rep2

Sample ID2 5

Result comment

Result state Not OK

Result state comment

Nominal value

Compliance state Released

Is result deleted No

Calculation formula R1=Q*C/m

Raw Results

Scope 1/1, Sample 1/4 (FTS2023GW-TS1NW1 8/16 11:00rep1; 2)

Sample (S1)

Creation time 2/22/2024 11:12:32 AM

Sample start time 2/22/2024 11:12:32 AM

Sample size 51.964 g

Correction factor 1.0

Temperature 25.0 °C

Density 1.0 g/mL

Titration stand (Titrationstand1)

Creation time 2/22/2024 11:12:32 AM

Measure (normal) [1] (Measurenormal1)

Creation time 2/22/2024 11:12:32 AM

T 24.7 °C

E 4.677 pH

tMe 01:12 min

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ESTMe 4.489 pH

Titration (EQP) [1] (TitrationEQP1)

Creation time 2/22/2024 11:13:45 AM
Terminated at Potential
EST 4.676 pH
VEX 0.56125 mL
VEND 0.56125 mL
t 11:11 min
QEND 0.056181 mmol
QEX 0.056181 mmol
EPD 4.670 pH
nEQ 0

Content (R1)

Creation time 2/22/2024 11:25:01 AM
Result NaN mmol/L
Decimal places 4
Substance None
M 1 g/mol
z 1

Rinse (Rinse1)

Creation time 2/22/2024 11:25:01 AM

Scope 1/1, Sample 2/4 (FTS2023GW-TS1NW1 8/16 11:00rep2; 3)

Sample (S1)

Creation time 2/22/2024 11:25:12 AM
Sample start time 2/22/2024 11:25:12 AM
Sample size 42.547 g
Correction factor 1.0
Temperature 25.0 °C
Density 1.0 g/mL

Titration stand (Titrationstand1)

Creation time 2/22/2024 11:25:12 AM

Measure (normal) [1] (Measurenormal1)

Creation time 2/22/2024 11:25:12 AM
T 25.1 °C
E 4.692 pH
tMe 01:12 min
ESTMe 3.640 pH

Titration (EQP) [1] (TitrationEQP1)

Creation time 2/22/2024 11:26:25 AM
Terminated at Potential
EST 4.689 pH
VEX 0.44850 mL
VEND 0.44850 mL

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t	10:10 min
QEND	0.044895 mmol
QEX	0.044895 mmol
EPD	4.662 pH
nEQ	0

Content (R1)

Creation time	2/22/2024 11:36:40 AM
Result	NaN mmol/L
Decimal places	4
Substance	None
M	1 g/mol
z	1

Rinse (Rinse1)

Creation time	2/22/2024 11:36:41 AM
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Scope 1/1, Sample 3/4 (FTS2023GW-T2NW1 8/16 12:00rep1; 4)

Sample (S1)

Creation time	2/22/2024 11:36:51 AM
Sample start time	2/22/2024 11:36:51 AM
Sample size	50.991 g
Correction factor	1.0
Temperature	25.0 °C
Density	1.0 g/mL

Titration stand (Titrationstand1)

Creation time	2/22/2024 11:36:52 AM
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Measure (normal) [1] (Measurenormal1)

Creation time	2/22/2024 11:36:52 AM
T	25.1 °C
E	5.052 pH
tMe	01:12 min
ESTMe	3.964 pH

Titration (EQP) [1] (TitrationEQP1)

Creation time	2/22/2024 11:38:04 AM
Terminated at	Potential
EST	5.053 pH
VEX	0.53325 mL
VEND	0.53325 mL
t	12:11 min
QEND	0.053378 mmol
QEX	0.053378 mmol
EPD	5.051 pH
nEQ	0

Content (R1)

Creation time	2/22/2024 11:50:21 AM
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Result	NaN mmol/L
Decimal places	4
Substance	None
M	1 g/mol
z	1

Rinse (Rinse1)

Creation time	2/22/2024 11:50:21 AM
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Scope 1/1, Sample 4/4 (FTS2023GW-T2NW1 8/16 12:00rep2; 5)

Sample (S1)

Creation time	2/22/2024 11:50:32 AM
Sample start time	2/22/2024 11:50:32 AM
Sample size	59.02 g
Correction factor	1.0
Temperature	25.0 °C
Density	1.0 g/mL

Titration stand (Titrationstand1)

Creation time	2/22/2024 11:50:32 AM
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Measure (normal) [1] (Measurenormal1)

Creation time	2/22/2024 11:50:33 AM
T	25.1 °C
E	5.041 pH
tMe	01:12 min
ESTMe	3.606 pH

Titration (EQP) [1] (TitrationEQP1)

Creation time	2/22/2024 11:51:45 AM
Terminated at	Potential
EST	5.041 pH
VEX	0.64675 mL
VEND	0.64675 mL
t	13:12 min
QEND	0.064740 mmol
QEX	0.064740 mmol
EPD	5.043 pH
nEQ	0

Content (R1)

Creation time	2/22/2024 12:05:03 PM
Result	NaN mmol/L
Decimal places	4
Substance	None
M	1 g/mol
z	1

Rinse (Rinse1)

Creation time	2/22/2024 12:05:03 PM
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After Sample Scopes

Park (Park1)

Creation time

2/22/2024 12:05:31 PM

Measure (normal) [1] Scope 1/1, Sample 1/4

Measured Values Scope 1/1, Sample 1/4

Time [s]	Meas. value [pH]	Temp. [°C]	dE [pH]
1	4.489	19.3	
2	4.629	19.9	0.140
3	4.656	20.7	0.027
4	4.669	21.5	0.013
5	4.678	22.2	0.009
6	4.684	22.8	0.006
7	4.688	23.2	0.004
8	4.691	23.5	0.003
9	4.693	23.8	0.002
10	4.694	24.0	0.001
11	4.694	24.2	0.000
12	4.694	24.3	0.000
13	4.694	24.4	0.000
14	4.694	24.4	0.000
15	4.693	24.5	-0.001
16	4.693	24.6	0.000
17	4.692	24.6	-0.001
18	4.692	24.6	0.000
19	4.691	24.6	-0.001
20	4.690	24.7	-0.001
21	4.690	24.7	0.000
22	4.689	24.7	-0.001
23	4.689	24.7	0.000
24	4.688	24.7	-0.001
25	4.688	24.7	0.000
26	4.687	24.7	-0.001
27	4.687	24.7	0.000
28	4.686	24.7	-0.001
29	4.686	24.7	0.000
30	4.685	24.7	-0.001
31	4.684	24.7	-0.001
32	4.682	24.7	-0.002
33	4.684	24.7	0.002
34	4.687	24.7	0.003
35	4.685	24.7	-0.002
36	4.683	24.7	-0.002
37	4.683	24.7	0.000
38	4.682	24.7	-0.001
39	4.682	24.7	0.000

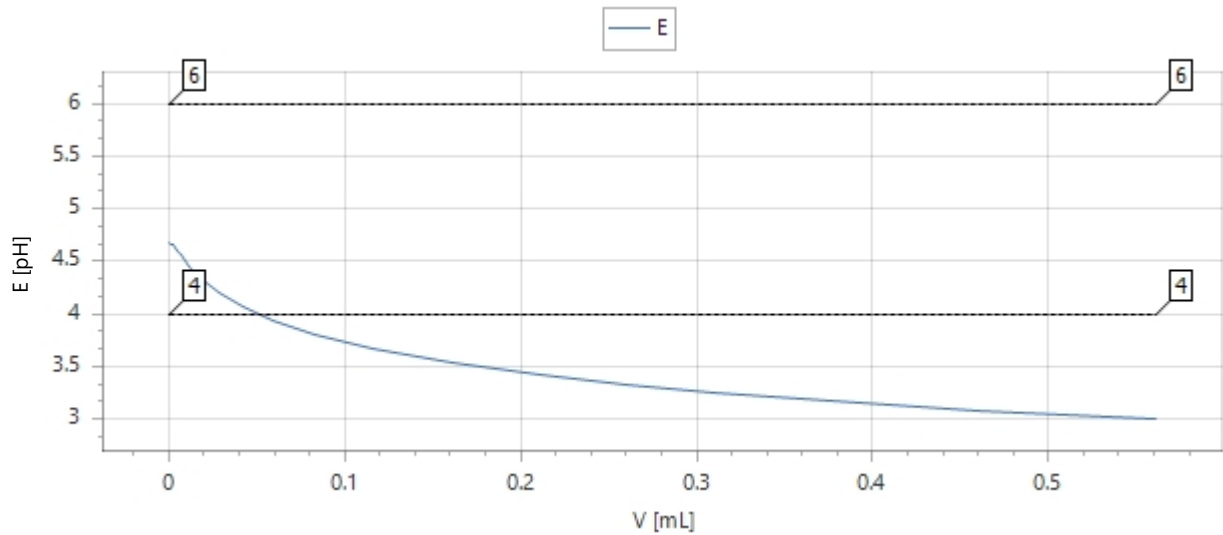
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40	4.681	24.7	-0.001
41	4.681	24.7	0.000
42	4.680	24.7	-0.001
43	4.681	24.7	0.001
44	4.681	24.7	0.000
45	4.681	24.7	0.000
46	4.680	24.7	-0.001
47	4.680	24.7	0.000
48	4.680	24.7	0.000
49	4.679	24.7	-0.001
50	4.679	24.7	0.000
51	4.678	24.7	-0.001
52	4.678	24.7	0.000
53	4.678	24.7	0.000
54	4.678	24.7	0.000
55	4.678	24.7	0.000
56	4.678	24.7	0.000
57	4.677	24.7	-0.001
58	4.677	24.7	0.000
59	4.677	24.7	0.000
60	4.677	24.7	0.000

Titration (EQP) [1] Scope 1/1, Sample 1/4

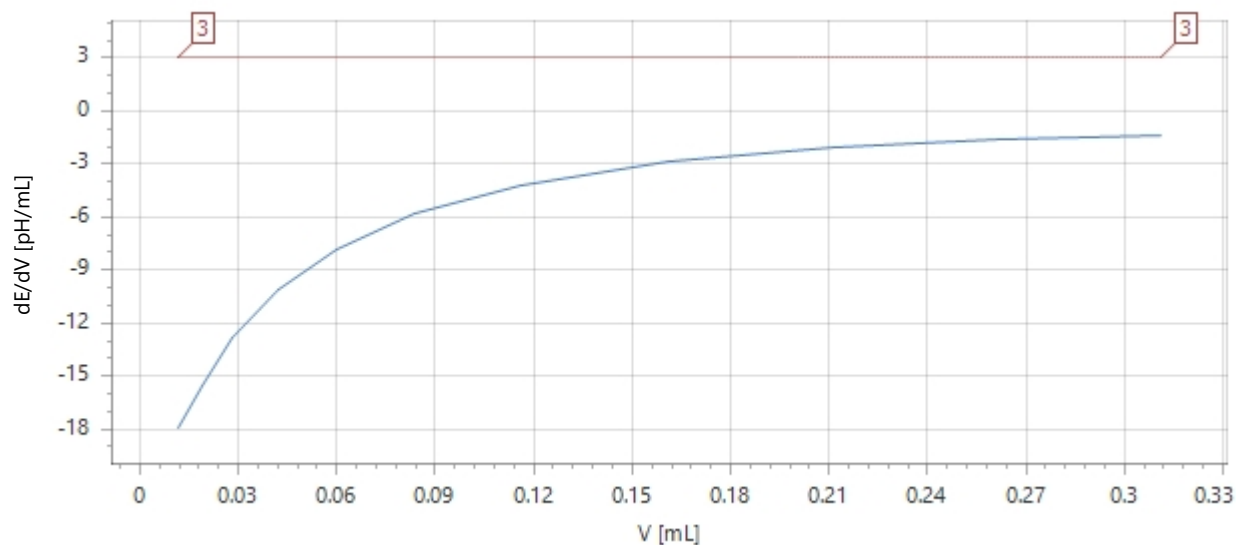
E - V Curve Scope 1/1, Sample 1/4



dE/dV - V Curve Scope 1/1, Sample 1/4

Titration report

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Measured Values Scope 1/1, Sample 1/4

Time [s]	V [mL]	Meas. value [pH]	dV [mL]	Temp. [°C]	dE [pH]	dE/dV [pH/mL]
0	0.00000	4.676		24.7		
91	0.00100	4.667	0.00100	24.8	-0.009	
121	0.00200	4.661	0.00100	24.8	-0.006	
151	0.00450	4.611	0.00250	24.8	-0.050	
181	0.00650	4.564	0.00200	24.8	-0.047	
212	0.01150	4.458	0.00500	24.8	-0.106	-17.88
242	0.01875	4.333	0.00725	24.8	-0.125	-15.50
272	0.02825	4.209	0.00950	24.8	-0.124	-12.87
302	0.04200	4.071	0.01375	24.8	-0.138	-10.11
332	0.05975	3.937	0.01775	24.8	-0.134	-7.80
363	0.08350	3.803	0.02375	24.9	-0.134	-5.80
393	0.11575	3.670	0.03225	24.9	-0.133	-4.18
423	0.16125	3.531	0.04550	24.9	-0.139	-2.86
454	0.21125	3.417	0.05000	24.9	-0.114	-2.07
484	0.26125	3.326	0.05000	24.9	-0.091	-1.63
514	0.31125	3.251	0.05000	24.9	-0.075	-1.35
544	0.36125	3.187	0.05000	24.9	-0.064	
574	0.41125	3.132	0.05000	24.9	-0.055	
605	0.46125	3.083	0.05000	24.9	-0.049	
635	0.51125	3.038	0.05000	24.9	-0.045	
665	0.56125	2.999	0.05000	24.9	-0.039	

Measure (normal) [1] Scope 1/1, Sample 2/4

Measured Values Scope 1/1, Sample 2/4

Time [s]	Meas. value [pH]	Temp. [°C]	dE [pH]
1	3.640	23.4	
2	4.649	23.5	1.009

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3	4.813	23.8	0.164
4	4.866	24.0	0.053
5	4.886	24.3	0.020
6	4.893	24.5	0.007
7	4.892	24.6	-0.001
8	4.887	24.7	-0.005
9	4.879	24.8	-0.008
10	4.870	24.9	-0.009
11	4.861	24.9	-0.009
12	4.852	25.0	-0.009
13	4.844	25.0	-0.008
14	4.835	25.0	-0.009
15	4.828	25.0	-0.007
16	4.820	25.0	-0.008
17	4.813	25.1	-0.007
18	4.806	25.1	-0.007
19	4.799	25.1	-0.007
20	4.793	25.1	-0.006
21	4.787	25.1	-0.006
22	4.782	25.1	-0.005
23	4.777	25.1	-0.005
24	4.772	25.1	-0.005
25	4.768	25.1	-0.004
26	4.764	25.1	-0.004
27	4.760	25.1	-0.004
28	4.756	25.1	-0.004
29	4.752	25.1	-0.004
30	4.749	25.1	-0.003
31	4.746	25.1	-0.003
32	4.742	25.1	-0.004
33	4.739	25.1	-0.003
34	4.736	25.1	-0.003
35	4.734	25.1	-0.002
36	4.731	25.1	-0.003
37	4.729	25.1	-0.002
38	4.726	25.1	-0.003
39	4.724	25.1	-0.002
40	4.722	25.1	-0.002
41	4.719	25.1	-0.003
42	4.717	25.1	-0.002
43	4.715	25.1	-0.002
44	4.714	25.1	-0.001
45	4.712	25.1	-0.002
46	4.710	25.1	-0.002
47	4.709	25.1	-0.001
48	4.707	25.1	-0.002
49	4.706	25.1	-0.001
50	4.704	25.1	-0.002
51	4.703	25.1	-0.001
52	4.701	25.1	-0.002
53	4.700	25.1	-0.001

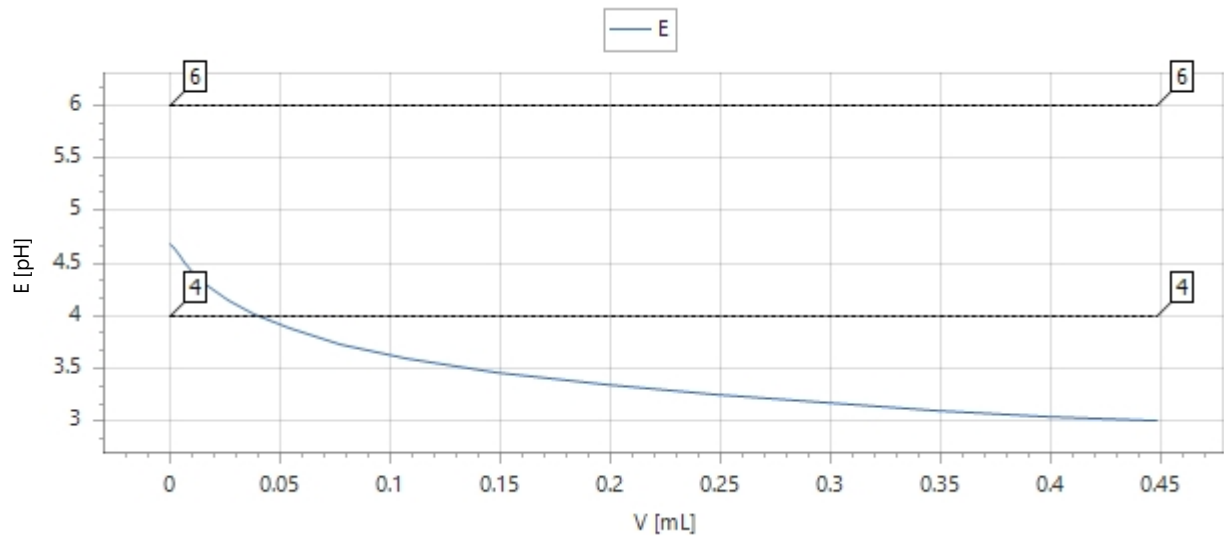
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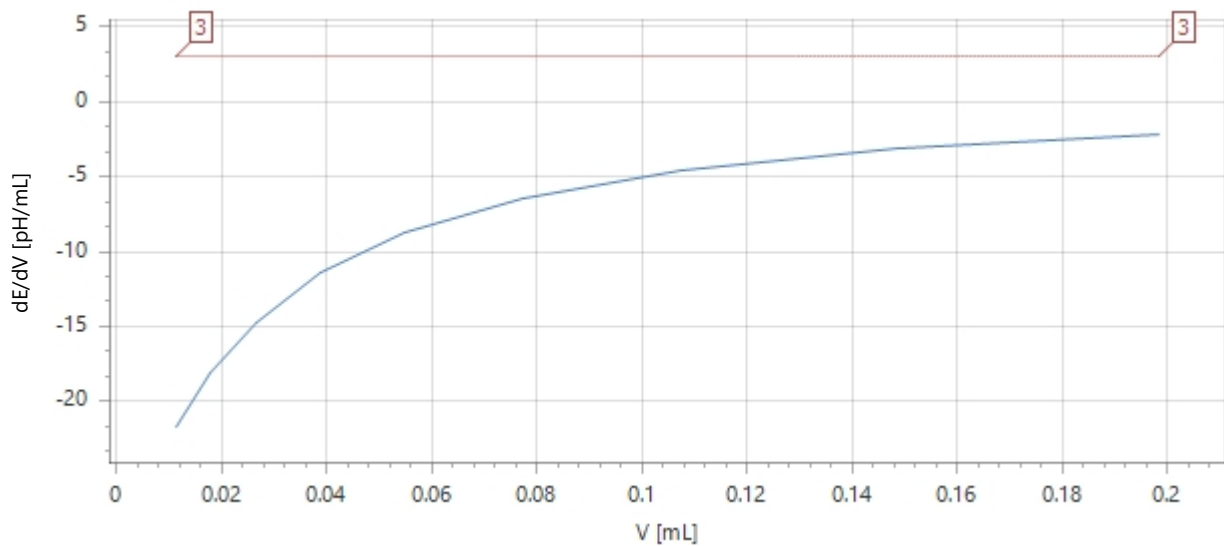
54	4.699	25.1	-0.001
55	4.698	25.1	-0.001
56	4.697	25.1	-0.001
57	4.695	25.1	-0.002
58	4.694	25.1	-0.001
59	4.693	25.1	-0.001
60	4.692	25.1	-0.001

Titration (EQP) [1] Scope 1/1, Sample 2/4

E - V Curve Scope 1/1, Sample 2/4



dE/dV - V Curve Scope 1/1, Sample 2/4



Measured Values Scope 1/1, Sample 2/4

Time [s]	V [mL]	Meas. value [pH]	dV [mL]	Temp. [°C]	dE [pH]	dE/dV [pH/mL]
0	0.00000	4.689		25.1		
91	0.00100	4.657	0.00100	25.1	-0.032	

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121	0.00200	4.646	0.00100	25.1	-0.011	
151	0.00450	4.576	0.00250	25.1	-0.070	
181	0.00650	4.518	0.00200	25.1	-0.058	
212	0.01125	4.401	0.00475	25.1	-0.117	-21.65
242	0.01775	4.274	0.00650	25.1	-0.127	-18.10
272	0.02650	4.145	0.00875	25.1	-0.129	-14.71
302	0.03875	4.007	0.01225	25.1	-0.138	-11.41
332	0.05475	3.876	0.01600	25.1	-0.131	-8.73
363	0.07725	3.738	0.02250	25.1	-0.138	-6.39
393	0.10725	3.604	0.03000	25.1	-0.134	-4.58
423	0.14850	3.467	0.04125	25.1	-0.137	-3.16
454	0.19850	3.346	0.05000	25.1	-0.121	-2.22
484	0.24850	3.251	0.05000	25.1	-0.095	
514	0.29850	3.173	0.05000	25.1	-0.078	
544	0.34850	3.108	0.05000	25.1	-0.065	
574	0.39850	3.048	0.05000	25.1	-0.060	
605	0.44850	3.000	0.05000	25.1	-0.048	

Measure (normal) [1] Scope 1/1, Sample 3/4

Measured Values Scope 1/1, Sample 3/4

Time [s]	Meas. value [pH]	Temp. [°C]	dE [pH]
1	3.964	23.6	
2	4.323	23.8	0.359
3	4.519	24.0	0.196
4	4.639	24.2	0.120
5	4.720	24.4	0.081
6	4.779	24.6	0.059
7	4.823	24.7	0.044
8	4.856	24.8	0.033
9	4.885	24.9	0.029
10	4.909	24.9	0.024
11	4.928	25.0	0.019
12	4.945	25.0	0.017
13	4.959	25.0	0.014
14	4.972	25.1	0.013
15	4.981	25.1	0.009
16	4.990	25.1	0.009
17	4.997	25.1	0.007
18	5.003	25.1	0.006
19	5.009	25.1	0.006
20	5.014	25.1	0.005
21	5.018	25.1	0.004
22	5.022	25.1	0.004
23	5.025	25.1	0.003
24	5.028	25.1	0.003
25	5.030	25.1	0.002
26	5.032	25.1	0.002
27	5.035	25.1	0.003

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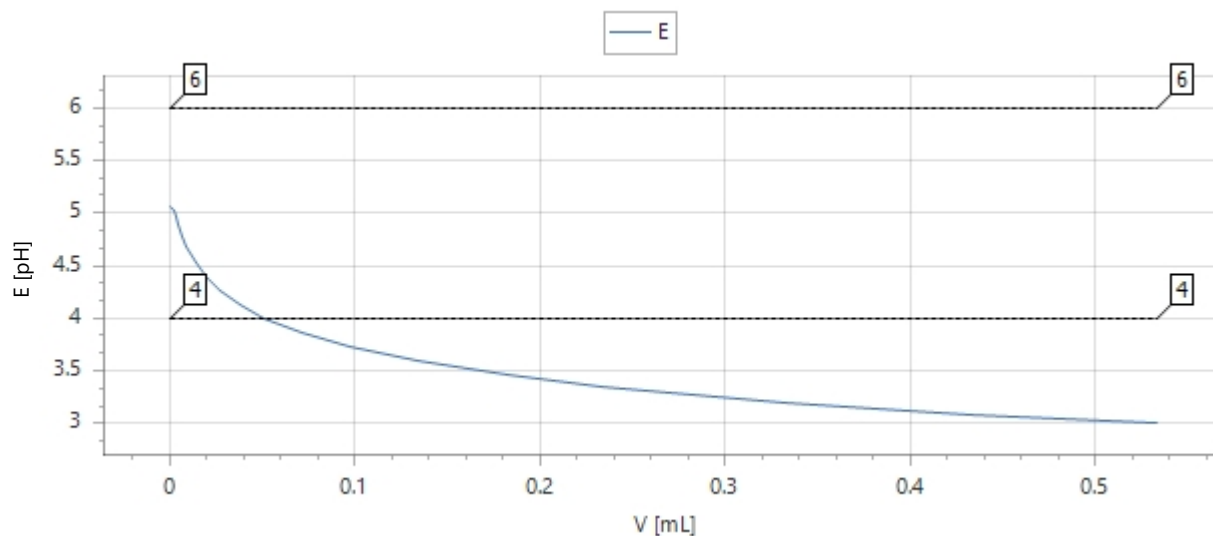
28	5.037	25.1	0.002
29	5.038	25.1	0.001
30	5.039	25.1	0.001
31	5.041	25.1	0.002
32	5.042	25.1	0.001
33	5.043	25.1	0.001
34	5.044	25.1	0.001
35	5.045	25.1	0.001
36	5.045	25.1	0.000
37	5.046	25.1	0.001
38	5.047	25.1	0.001
39	5.047	25.1	0.000
40	5.048	25.1	0.001
41	5.049	25.1	0.001
42	5.049	25.1	0.000
43	5.049	25.1	0.000
44	5.050	25.1	0.001
45	5.050	25.1	0.000
46	5.050	25.1	0.000
47	5.051	25.1	0.001
48	5.051	25.1	0.000
49	5.051	25.1	0.000
50	5.051	25.1	0.000
51	5.051	25.1	0.000
52	5.052	25.1	0.001
53	5.052	25.1	0.000
54	5.052	25.1	0.000
55	5.052	25.1	0.000
56	5.052	25.1	0.000
57	5.052	25.1	0.000
58	5.052	25.1	0.000
59	5.052	25.1	0.000
60	5.052	25.1	0.000

Titration (EQP) [1] Scope 1/1, Sample 3/4

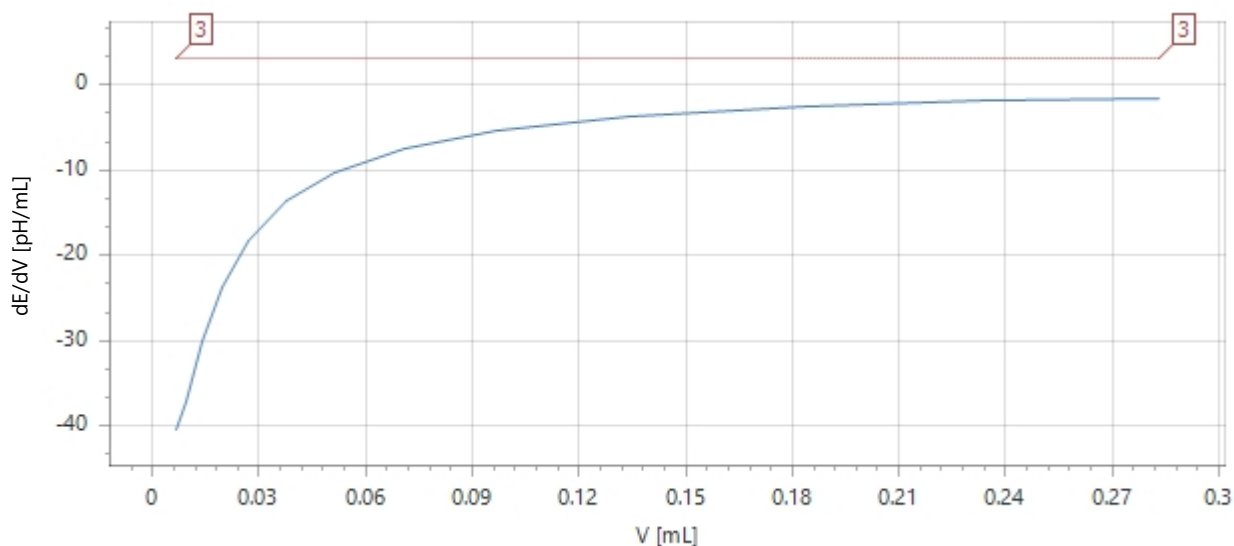
E - V Curve Scope 1/1, Sample 3/4

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dE/dV - V Curve Scope 1/1, Sample 3/4



Measured Values Scope 1/1, Sample 3/4

Time [s]	V [mL]	Meas. value [pH]	dV [mL]	Temp. [°C]	dE [pH]	dE/dV [pH/mL]
0	0.00000	5.053		25.1		
91	0.00100	5.049	0.00100	25.1	-0.004	
121	0.00200	5.025	0.00100	25.1	-0.024	
151	0.00300	4.971	0.00100	25.1	-0.054	
181	0.00425	4.899	0.00125	25.1	-0.072	
212	0.00650	4.782	0.00225	25.1	-0.117	-40.31
242	0.00925	4.675	0.00275	25.1	-0.107	-37.20
272	0.01400	4.529	0.00475	25.1	-0.146	-30.00
302	0.01950	4.400	0.00550	25.1	-0.129	-23.79
332	0.02700	4.271	0.00750	25.1	-0.129	-18.37
363	0.03775	4.129	0.01075	25.1	-0.142	-13.70
393	0.05125	4.002	0.01350	25.1	-0.127	-10.30
423	0.07100	3.862	0.01975	25.1	-0.140	-7.46
454	0.09700	3.730	0.02600	25.1	-0.132	-5.38

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484	0.13375	3.592	0.03675	25.1	-0.138	-3.72
514	0.18325	3.457	0.04950	25.1	-0.135	-2.51
544	0.23325	3.353	0.05000	25.1	-0.104	-1.86
574	0.28325	3.271	0.05000	25.1	-0.082	-1.49
605	0.33325	3.201	0.05000	25.0	-0.070	
635	0.38325	3.141	0.05000	25.0	-0.060	
665	0.43325	3.089	0.05000	25.0	-0.052	
695	0.48325	3.042	0.05000	25.0	-0.047	
726	0.53325	3.000	0.05000	25.0	-0.042	

Measure (normal) [1] Scope 1/1, Sample 4/4

Measured Values Scope 1/1, Sample 4/4

Time [s]	Meas. value [pH]	Temp. [°C]	dE [pH]
1	3.606	23.7	
2	4.164	23.8	0.558
3	4.394	24.0	0.230
4	4.531	24.3	0.137
5	4.624	24.4	0.093
6	4.696	24.6	0.072
7	4.748	24.7	0.052
8	4.787	24.8	0.039
9	4.823	24.9	0.036
10	4.853	24.9	0.030
11	4.878	25.0	0.025
12	4.899	25.0	0.021
13	4.915	25.0	0.016
14	4.930	25.1	0.015
15	4.943	25.1	0.013
16	4.953	25.1	0.010
17	4.962	25.1	0.009
18	4.971	25.1	0.009
19	4.978	25.1	0.007
20	4.984	25.1	0.006
21	4.990	25.1	0.006
22	4.995	25.1	0.005
23	4.999	25.1	0.004
24	5.004	25.1	0.005
25	5.007	25.1	0.003
26	5.010	25.1	0.003
27	5.013	25.1	0.003
28	5.015	25.1	0.002
29	5.017	25.1	0.002
30	5.019	25.1	0.002
31	5.021	25.1	0.002
32	5.023	25.1	0.002
33	5.024	25.1	0.001
34	5.026	25.1	0.002
35	5.027	25.1	0.001

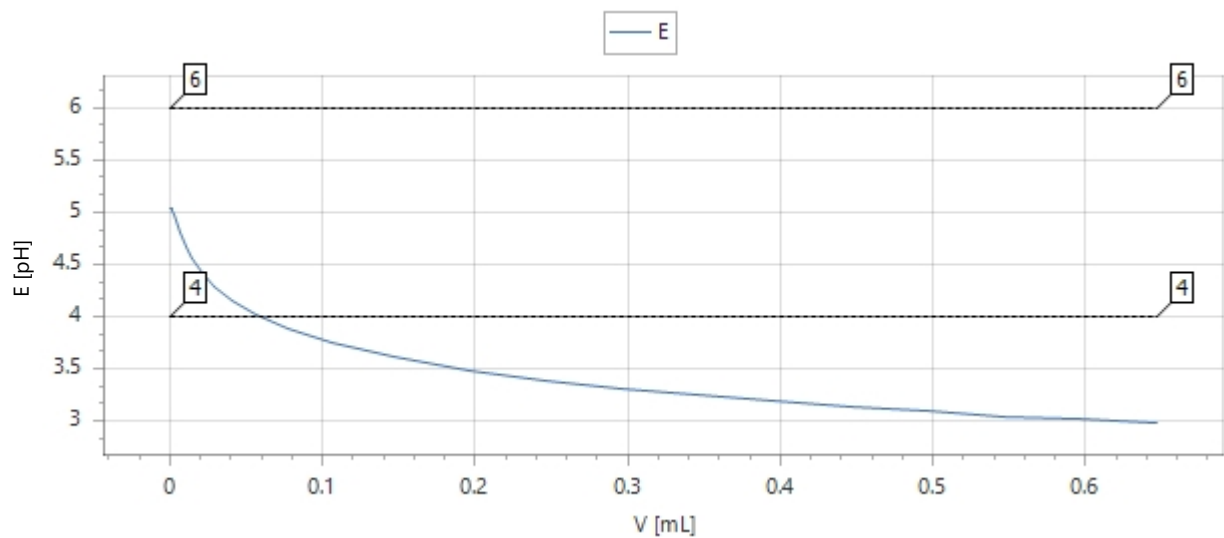
Titration report

LabX

36	5.028	25.1	0.001
37	5.029	25.1	0.001
38	5.030	25.1	0.001
39	5.031	25.1	0.001
40	5.032	25.1	0.001
41	5.033	25.1	0.001
42	5.034	25.1	0.001
43	5.034	25.1	0.000
44	5.035	25.1	0.001
45	5.035	25.1	0.000
46	5.036	25.1	0.001
47	5.036	25.1	0.000
48	5.037	25.1	0.001
49	5.037	25.1	0.000
50	5.038	25.1	0.001
51	5.038	25.1	0.000
52	5.038	25.1	0.000
53	5.039	25.1	0.001
54	5.039	25.1	0.000
55	5.039	25.1	0.000
56	5.040	25.1	0.001
57	5.040	25.1	0.000
58	5.040	25.1	0.000
59	5.041	25.1	0.001
60	5.041	25.1	0.000

Titration (EQP) [1] Scope 1/1, Sample 4/4

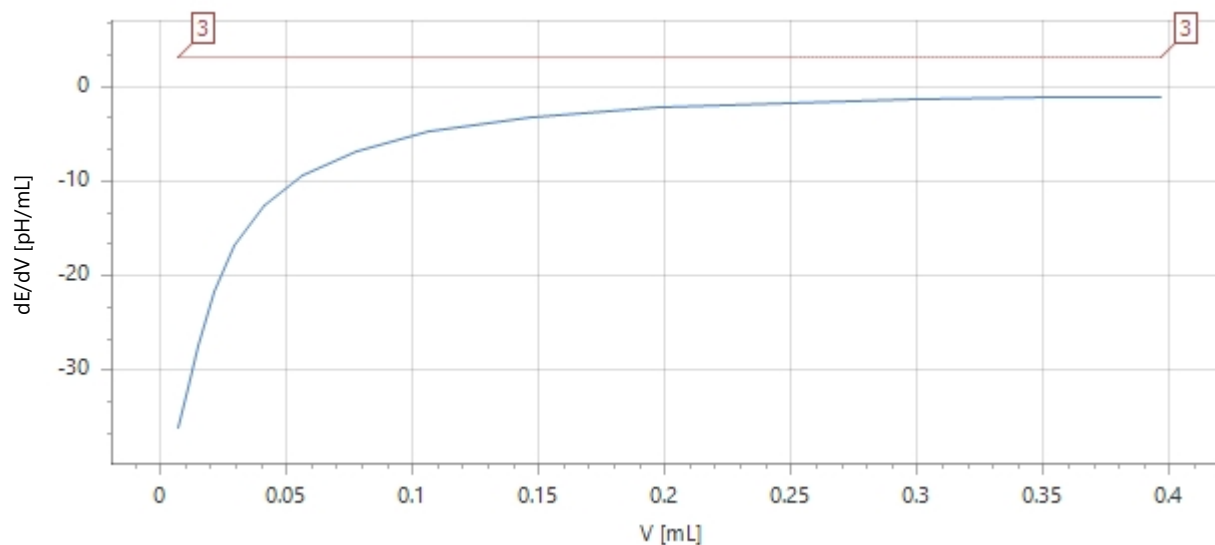
E - V Curve Scope 1/1, Sample 4/4



dE/dV - V Curve Scope 1/1, Sample 4/4

Titration report

LabX



Measured Values Scope 1/1, Sample 4/4

Time [s]	V [mL]	Meas. value [pH]	dV [mL]	Temp. [°C]	dE [pH]	dE/dV [pH/mL]
0	0.00000	5.041		25.1		
91	0.00100	5.042	0.00100	25.1	0.001	
121	0.00200	5.023	0.00100	25.1	-0.019	
151	0.00300	4.977	0.00100	25.1	-0.046	
181	0.00425	4.915	0.00125	25.1	-0.062	
212	0.00675	4.804	0.00250	25.1	-0.111	-35.95
242	0.01025	4.676	0.00350	25.1	-0.128	-32.74
272	0.01475	4.550	0.00450	25.0	-0.126	-27.37
302	0.02100	4.416	0.00625	25.0	-0.134	-21.64
332	0.02925	4.286	0.00825	25.0	-0.130	-16.75
363	0.04075	4.150	0.01150	25.0	-0.136	-12.54
393	0.05600	4.017	0.01525	25.0	-0.133	-9.30
423	0.07750	3.880	0.02150	25.0	-0.137	-6.75
454	0.10650	3.746	0.02900	25.0	-0.134	-4.80
484	0.14675	3.609	0.04025	25.0	-0.137	-3.30
514	0.19675	3.484	0.05000	25.0	-0.125	-2.28
544	0.24675	3.388	0.05000	25.0	-0.096	-1.73
574	0.29675	3.308	0.05000	25.0	-0.080	-1.41
605	0.34675	3.242	0.05000	25.0	-0.066	-1.20
635	0.39675	3.185	0.05000	25.0	-0.057	-1.06
665	0.44675	3.134	0.05000	25.0	-0.051	
695	0.49675	3.089	0.05000	25.0	-0.045	
726	0.54675	3.048	0.05000	25.0	-0.041	
756	0.59675	3.011	0.05000	25.0	-0.037	
786	0.64675	2.977	0.05000	25.0	-0.034	

Used Instruments

Name SERCtitrate Serial number B433917554

Titration report



Responsible User

Time	Login name	Full name
2/22/2024 11:12:30 AM	SystemInternal	System internal account